TETHER

KO/PEP vs AAPL/MSFT: when correlation is not a book

TETHER RelVal desk · walk-forward research note

This note is the recruiter-facing record of what the desk actually tests. The live app screens a sector, FDR-c

1. What the desk is allowed to know

Walk-forward split: **discover on or before 2024-01-01, trade after**. Static OLS hedge, Engle–Granger p-value

That is the difference between a pairs tutorial and a book. In-sample Sharpe is the in-sample story. Out-of-sa

2. Multiple testing

A five-name sector has ten unique pairs. Ten Engle–Granger tests at 5% will produce false “cointegrated” hits.

- raw Engle–Granger p
- **Benjamini–Hochberg FDR**
- Bonferroni

The sentence the desk prints is the one that matters: *N of K pairs survive FDR. D of T TETHERED names die liv

3. Negative control: AAPL / MSFT

AAPL and MSFT share a tech factor. They often **move together**. That is not a mean-reverting spread.

On a recent full-sample run the desk labeled the pair **BROKEN** (Tether Score ~25) with a ~**114-day** Ornste

Rolling Engle–Granger p-values that sit above 0.05 are the tether decaying in public. Treat the residual as **

4. Working hypothesis: KO / PEP

Coca-Cola and PepsiCo are the textbook consumer-staples pair: similar cost bases, overlapping distribution, no
Read the Desk table:

| Sample | What it is allowed to do |

|---|---|

| In-sample | Estimate cointegration, freeze β, score the tether |

| Out-of-sample | Trade z-score rules with that frozen hedge; re-score the tether |

If KO/PEP’s OOS Sharpe is below IS Sharpe, **that result stays on the Desk**. A pretty equity curve that canno
One blotter line is enough to explain the clock: side, entry z, days held, holding time vs half-life, dollar P

5. Residual risk the fund actually charges you for

Borrow. Short inventory is not free. The toggle charges an annualized borrow rate (default 50 bps) on shor
Net book beta. Leg betas vs SPY are not the book. Dollar holdings times those betas, divided by capital, i

6. How to read TETHER in 40 seconds

1. **Universe** — Consumer Staples or Energy. Scan. Count FDR survivors. Load KO/PEP, not AAPL/MSFT.
2. **Desk** — Walk-forward table. If OOS Sharpe is worse, leave it. Ask Research: *why is this BROKEN?*
3. **Tape** — Spread vs equilibrium, z-bands, trade marks, PnL vs pair buy-and-hold.

The engine is in `src/logic.py`. This note is what you send instead of the file.

7. What would change the read

• A structural break (product mix, merger, index reconstitution) that the rolling p-value has not yet printed.
• Borrow spikes that the 50 bp assumption understates.
• A Kalman hedge that lets net β wander through a factor crisis.

TETHER’s job is not to promise that pairs always pay. It is to show **when the tether is still holding, and wh